

Humanoid Robot Using Edge AI



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Final Year B-Tech Project Presentation, 8th Semester,
Review-2

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Objective

- **Project Goal:** Minimize the major aspects of an integrated robot by switching to Edge AI.
- **Key Results:** This approach leads to reduced computational and hardware size and includes a sustainable battery solution.
- **Future Impact:** This new approach paves the way for the future of remote and isolated terrain-oriented robots for navigation and for household help bots.

Key features:-

- Camera for vision
- Manipulator (ARM)
- Mobility (wheeled structure)
- Speech enable (sensors and speakers)
- Edge ai (software)

Introduction

Motivation

The motivation for this project stems from the rapid advancement of autonomous systems and the increasing need for robots that can navigate human-centric environments safely. Traditional robotics often rely on heavy, cloud-dependent processing. This project is motivated by the challenge of implementing Edge AI—performing complex computer vision and decision-making locally on a Raspberry Pi—to minimize latency and improve privacy. Building a humanoid form factor, even a wheeled one, provides a more natural interface for human interaction compared to standard rovers.

Introduction

OBJECTIVES

The primary objectives of this project are:

To deploy an optimized YOLOv8 model on a Raspberry Pi 4/5 capable of identifying and locating humans in real-time.

To build a ROS 2-based software architecture that modularizes vision, tracking logic, and hardware communication.

To implement a tracking algorithm that maintains a target human at the center of the robot's field of vision using precise motor control.

To establish reliable cross-platform communication (Serial/USB) for real-time command execution on an ESP32 slave device.

Literature Survey

S.No.	Inferences	References
1.	"Edge computing can cut latency by up to 75% compared to cloud solutions, with advanced setups reducing data processing delays by about 67%.."	(Source: Konakanchi, S. (2025). EDGE COMPUTING FOR LATENCY-SENSITIVE AI APPLICATIONS. International Journal of Civil Engineering and Technology, 16(2), 2038–2046.)
2.	"Redundant parameters in DNNs boost learning but strain edge devices with limited power and memory; techniques like pruning and quantization address this".	(Source: Elawady, M., et al. (2025). A survey of model compression techniques: past, present, and future. Frontiers in Robotics and AI, 12.)
3.	"Edge AI processes data locally to reduce latency and enhance privacy, using methods like federated learning, differential privacy, and SMPC".	(Source: Kumar, B. (2025). EDGE AI FOR PRIVACY-PRESERVING DATA ANALYTICS IN IOT-ENABLED SMART CITIES. ResearchGate)
4.	"Resource constraints on edge devices hinder the deployment of deep learning models, especially during inference. Intensive inference tasks can rapidly deplete battery power and cause overheating."	.(Source: Ganesan, N., et al. (2025). Edge AI: Challenges and Opportunities in Real-Time Processing. ResearchGate)

Literature Survey

S.No.	Inferences	References
1.	"IMU–model fusion enables real-time force and moment estimation without physical sensors. Running at 200 Hz with <5% error, it greatly enhances balance stability in dynamic environments."	Li, W., Li, Y., Ma, Z., et al. (2025). Research on Balance Control of Humanoid Robot Based on Inertial Measurement Unit. Proceedings of the 2025 8th International Conference on Advanced Control and Emergent Systems (ACES). EWA Direct.
2.	"Audio-visual integration enables the robot to attend to speaking or waving persons, while multimodal sensing increases robustness in perceiving and interpreting its surroundings."	Lee, H., Yun, H., Kim, M., et al. (2015). Audio-visual integration for human-robot interaction in multi-person scenarios. 2015 15th International Conference on Control, Automation and Systems (ICCAS). IEEE.
3.	"LiDAR delivers accurate 3D sensing for robotic navigation, and when paired with SLAM, allows real-time mapping and localization."	Juhani, H., & Lehtonen, M. (2024). Study of Human–Robot Interactions for Assistive Robots Using Machine Learning and Sensor Fusion Technologies. Electronics, 13(16), 3285. MDPI.

Structure Inspiration

Table 1 continued from previous page

Name & Company	Picture	Humanlike features	Humanlike functions	AI & technical functions	Applications	Shortage
Phoenix ³⁰ , Sanctuary AI, Canada		Non-humanlike head, face, neck, torso, arms, hands, fingers, legs, feet	Mobility, walking, fine manipulation and dexterity, teleoperation, telepresence, touch/haptic communication, grasping, autonomous	General-purpose robot, path planning, navigation, object detection, LLMs for language-to-action	Manufacturing, retail	No facial expressions
Promobot ³¹ , Promobot, Europe		Non-humanlike head, neck, torso, arms, hands, movement platform	Conversation, body movement, interaction, navigation	Face and speech recognition, path planning, navigation, object detection, interaction, autonomous; Ubutu, compatible with third-party software	Service robot, consultant, promoter, concierge, tour guide, assistant	Not bipedal, no facial expressions, limited advanced AI capabilities
RT2 ³² , Google, US		Non-humanlike structure	Perception, learning, reasoning, control	Human recognition, LLMs, vision-language-vision learning from web and robot, learning to control, multi-task demonstration, reasoning, symbol understanding	Research, grasping, retail, open-world manipulation	No human features
Sophia ³³ , Hanson Robotics, China		Head, anthropomorphic face, skin, neck, torso, arms, hands, fingers, legs, feet	Conversation, gesture, facial expression, interaction, walking, emotion	Human-robot interaction, object detection, symbolic AI, perception, NLP, expert systems, adaptive motor control, generation	Research, demonstration, medicine, education, service, entertainment, human-robot interaction	Scripted interaction, limited physical mobility

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Structure inspiration



**Promobot
Structure**

Promobot Humanoid Robot:

A service-oriented humanoid robot designed to resemble and communicate like a human, featuring a realistic face, articulated arms, speech capability, and autonomous wheeled movement. Its structure includes mechanical, electronic, and AI systems integrated to perform human-like social interactions in real-world environments.

Structure :

- Head
- Torso
- Arms
- Mobility base(wheel base moment)
- Sensor system
- Power and control unit

Structure inspiration

Q. Why is Promobot considered as humanoid Robot even though it moves on wheels?

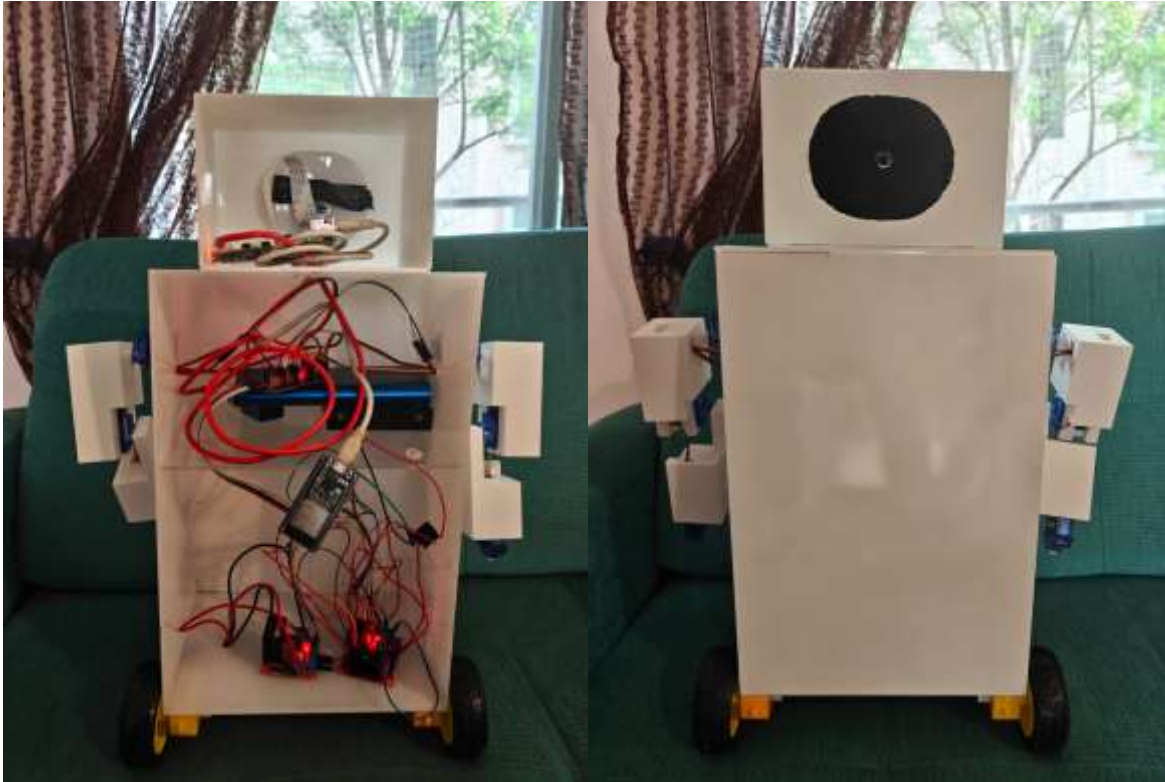
Ans. Promobot is called a humanoid robot because it *looks and behaves like a human* (face, voice, expressions, gestures), even though it rolls on wheels instead of walking on legs.



**Promobot
Structure**

- It **resembles the human body** (face, torso, arms).
- It **behaves and communicates like a human** (speech, facial expressions, gestures).
- It is **designed for human–robot social interaction**, not just industrial or technical tasks.
- Its mobility system (wheels) is a *functional choice*, not a defining factor — the humanoid nature comes from its **form and function**, not how it moves.

Proposed work



This project presents a vision-guided humanoid robot designed for autonomous human tracking using a Master-Slave architecture. The system utilizes a Raspberry Pi as the primary vision node, running Ubuntu 22.04 for stability. Real-time object detection is achieved via YOLOv8n, processing input from a USB webcam to generate precise tracking coordinates.

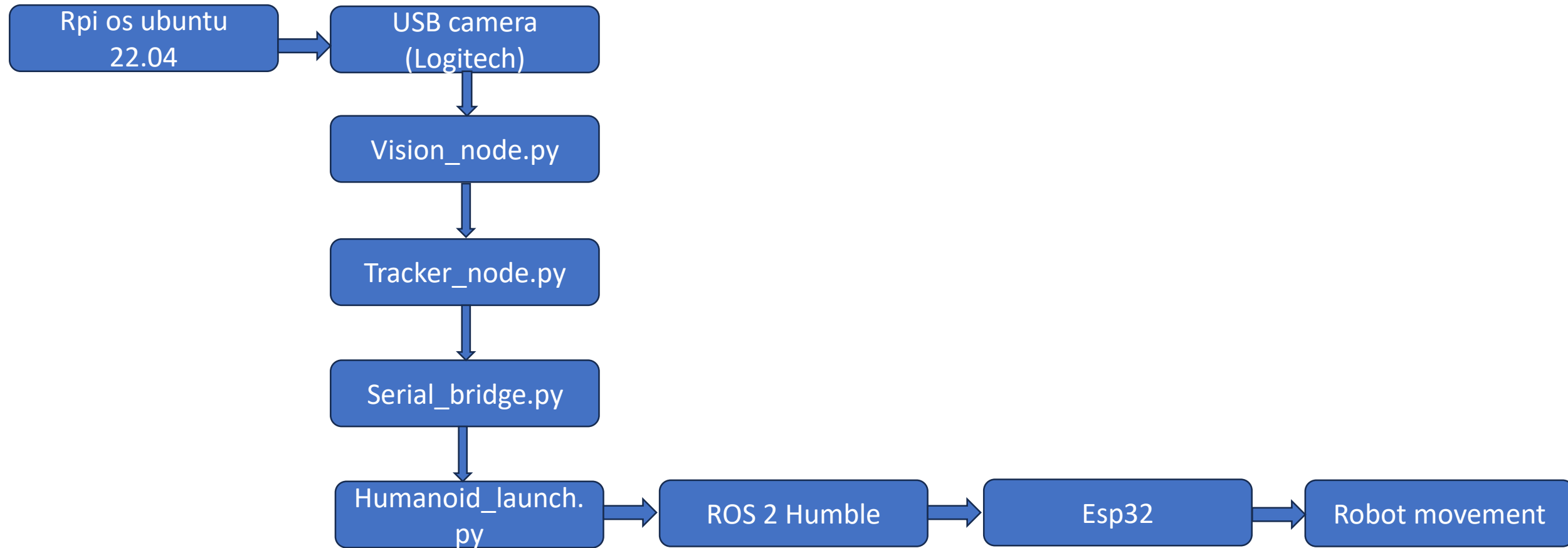
These instructions are transmitted to an ESP32 microcontroller, which executes PID control logic to manage motor torque and acceleration. Weighing 2.5kg, the prototype integrates Edge AI and ROS concepts to bridge the gap between high-level computer vision and low-level hardware actuation.

Hardware

Module	Component / Option	Notes
Processing Unit (“Brain”)	Raspberry Pi 4b 4gb	Full AI board, GPU-enabled, good for moderate Edge AI tasks.
Vision module	USB camera(Logitech)	Captures images and video frames
Motors / Actuators / Sensors	Servos & stepper motors L298n motor driver	Reliable actuation; include a few extra for replacements. .
Power / Battery	Lithium-ion pack, power bank	Size according to maximum load; consider energy optimization.

methodology

Flow chart: Raspberry pi os Directory flow



Software

AI/ML Frameworks (The Core)

- TensorFlow Lite (TFLite / LiteRT):
Includes a TFLite Converter to optimize and shrink your model for edge devices (e.g., quantization to reduce model size).
- ONNX Runtime:
A high-performance inference engine that supports models converted from various frameworks (like PyTorch or TensorFlow) into the Open Neural Network Exchange (ONNX) format.

ROS 2

ROS 2 HUMBLE is used for pipelining and launching the nodes all together. Used to fetch post processing data and send it to ESP32 for motor controls.

Software

Arduino IDE

- The Arduino handles the real-time, low-level tasks like reading motor encoders, controlling servos, and reading simple sensors.

RASPBERRY PI OS

- Raspberry Pi os – UBUNTU 22.04

Serial Communication Library

- A Python library like pyserial is essential for the Raspberry Pi to communicate with and send commands to the Arduino via the USB cable or GPIO pins.

AutoCAD

- AutoCAD is used to design the humanoid robot's structural framework with precise dimensions and geometric accuracy. It helps visualize and refine the robot's body parts, joints, and mechanical components before fabrication.

Results

```
#140 (30.01 fps) exp 33175.00 ag 4.00 dg 1.00
#141 (30.01 fps) exp 33251.00 ag 4.27 dg 1.00
#142 (30.01 fps) exp 33251.00 ag 4.41 dg 1.01
#143 (30.01 fps) exp 33251.00 ag 4.27 dg 1.02
#144 (30.01 fps) exp 33251.00 ag 4.06 dg 1.01
#145 (30.01 fps) exp 32816.00 ag 4.00 dg 1.00
#146 (30.00 fps) exp 32816.00 ag 4.00 dg 1.00
#147 (30.00 fps) exp 33251.00 ag 4.13 dg 1.00
Mode selection for 3280x2464:12:P
SRGBB10_CSI2P,640x480/0 - Score: 10248.8
SRGBB10_CSI2P,1640x1232/0 - Score: 6744
SRGBB10_CSI2P,1920x1080/0 - Score: 6655.48
SRGBB10_CSI2P,3280x2464/0 - Score: 1000
SRGBB8,640x480/0 - Score: 11248.8
SRGBB8,1640x1232/0 - Score: 7744
SRGBB8,1920x1080/0 - Score: 7655.48
SRGBB8,3280x2464/0 - Score: 2000
[0:12:24.994583724] [1901] INFO Camera camera.cpp:1215 configuring streams: (0)
3280x2464-YUV420/sYCC (1) 3280x2464-SBGGR10_CSI2P/RAW
[0:12:24.995406163] [1904] INFO RPI rasp.cpp:620 Sensor: /base/soc/l2c0mux/l2c@1
/imx219@10 - Selected sensor format: 3280x2464-SBGGR10_1X10/RAW - Selected unica
n format: 3280x2464-pBAA/RAW
Still capture image received
pi4@raspberrypi:~$
```

```
114 char c = Serial.read();
115
116 Serial.print(c); // debug incoming data
```

Output Serial Monitor X

Message (Enter to send message to 'ESP32 Dev Module' on '/dev/ttyUSB0')

```
Test done
Test done
Test done
Test done
Test done
F
→ received
CMD: F
B
→ received
CMD: B
```

Parameter	Metric	Technical Significance
Average Frame Rate	30.01 FPS	Provides 33.3ms updates to the Vision Node.
Exposure Time	33175.00 ns	Optimized to prevent motion blur during robot rotation.
Analogue Gain	4.00	Balanced sensitivity for standard laboratory lighting.
Digital Gain	1.00	Minimal digital noise, preserving edge detail for YOLOv8.

This figure shows the testing successful for communication between RPI and ESP32

Calculations

For the selected human target, the bounding box parameters are extracted as:

$$B = (x, y, w, h) \quad (1)$$

where x and y represent the top-left corner coordinates of the bounding box, and w and h denote the width and height of the detected region respectively.

The center of the detected person is computed as:

$$c_x = x + \frac{w}{2}, \quad c_y = y + \frac{h}{2} \quad (2)$$

These center coordinates are used by the tracking controller to align the robot with the target.

To determine whether the robot should turn left or right, the horizontal offset of the target center is calculated relative to the image center.

Let:

- x_{center} = horizontal center of detected person
- W_{img} = image width

The horizontal tracking error is defined as:

$$e_x = x_{center} - \frac{W_{img}}{2} \quad (3)$$

where $\frac{W_{img}}{2}$ represents the camera center.

The robot turning logic is:

- **L (Left):** if $e_x < -T_x$
- **R (Right):** if $e_x > T_x$

where T_x is the predefined horizontal threshold.

Calculations

Since the robot follows the target, distance is estimated by comparing the current bounding box area with a calibrated reference area corresponding to the desired following distance.

The current target area is:

$$A_{curr} = width \times height \quad (4)$$

where *width* and *height* are the dimensions of the detected bounding box.

Let:

- A_{ref} = reference area at desired distance
- M = allowable margin

The movement logic is:

- **F (Forward):** if $A_{curr} < (A_{ref} - M)$
- **B (Backward):** if $A_{curr} > (A_{ref} + M)$

Thus, when the detected person appears smaller, the robot moves forward, and when the person appears too close, the robot moves backward.

The robot remains stationary when the target is already aligned and positioned within the desired following range.

The stop command is triggered when:

- $|e_x| \leq T_x$
- $|A_{curr} - A_{ref}| \leq M$

or when no person is detected in the frame.

Hence:

$$S = \text{Stop} \quad (5)$$

under dead-zone conditions.

The final command set:

$$\{L, R, F, B, S\} \quad (6)$$

Conclusion

The presentation effectively concluded that Edge AI is a critical solution for humanoid robots. By shifting processing to the edge device, it overcomes

- cloud dependency
- enabling reduced hardware size
- sustainable power

and real-time functionality for the next generation of intelligent bots.

Future work

The proposed humanoid robot establishes a strong foundation for intelligent assistive and service robotics. In future developments, several advanced capabilities can be integrated to improve autonomy, interaction quality, and cognitive intelligence.

- **Obstacle Avoidance:** The robot can be upgraded with ultrasonic sensors, LiDAR, or depth cameras to detect static and dynamic obstacles in real time. This enhancement would enable safer navigation in crowded and unknown environments.
- **Conversational Chatbot Using Gemma 3:** An onboard chatbot system based on the Gemma 3 language model can be integrated for intelligent natural language interaction. This would allow the robot to answer queries, provide assistance, and maintain contextual conversations without cloud dependency.
- **Real-Time Speech Interaction:** Speech-to-text and text-to-speech modules can be added to enable continuous real-time talking capability. This would make communication more natural and user friendly in domestic and public environments.

Future work

- **Voice Command Control:** The robot can be enhanced with voice-triggered command recognition for tasks such as movement control, follow mode activation, navigation requests, and service assistance.
- **Preset Arm Movements:** Programmable arm gestures such as waving, greeting, object pointing, handshaking, and assistive motion patterns can be incorporated to improve human-robot interaction.
- **Fully Functional Talking Enabled Edge AI Neural Cognitive User Bot:** A long-term objective is to transform the system into a fully functional talking humanoid assistant with onboard neural cognition. Such a robot would combine perception, memory, speech intelligence, reasoning, emotional interaction, and autonomous decision making while operating entirely on edge hardware.

These future enhancements would significantly expand the practical usability of the proposed platform in healthcare, education, domestic assistance, security, and smart companion robotics.

References

1. (Source: Konakanchi, S. (2025). EDGE COMPUTING FOR LATENCY-SENSITIVE AI APPLICATIONS. International Journal of Civil Engineering and Technology, 16(2), 2038–2046.)
2. (Source: Elawady, M., et al. (2025). A survey of model compression techniques: past, present, and future. Frontiers in Robotics and AI, 12.)
3. (Source: Kumar, B. (2025). EDGE AI FOR PRIVACY-PRESERVING DATA ANALYTICS IN IOT-ENABLED SMART CITIES. ResearchGate)
4. ."(Source: Ganesan, N., et al. (2025). Edge AI: Challenges and Opportunities in Real-Time Processing. ResearchGate)
5. Li, W., Li, Y., Ma, Z., et al. (2025). Research on Balance Control of Humanoid Robot Based on Inertial Measurement Unit. Proceedings of the 2025 8th International Conference on Advanced Control and Emergent Systems (ACES). EWA Direct.
6. Lee, H., Yun, H., Kim, M., et al. (2015). Audio-visual integration for human-robot interaction in multi-person scenarios. 2015 15th International Conference on Control, Automation and Systems (ICCAS). IEEE.
7. uhani, H., & Lehtonen, M. (2024). Study of Human–Robot Interactions for Assistive Robots Using Machine Learning and Sensor Fusion Technologies. Electronics, 13(16), 3285. MDPI.

Thank You